

Docker: A Developer's Best Friend



Knowledge of Docker is a prerequisite for building scalable and portable applications in the cloud era. Let's explore what Docker is, how it works, why it is so popular, and when you should avoid using it.

Agility, scalability, and consistency of software systems are musts in the AI world. As these systems grow in complexity, there is more pressure on IT teams to produce, test, and implement applications quickly without compromising on their reliability or performance. Traditional deployment methods, such as virtual machines or manual configuration, fall short in performance today. Docker's entry into this field has changed the face of software development and deployment.

Docker is an open source platform developed to help automate the deployment of applications inside lightweight, portable-running containers. These containers house anything the application needs to run -- code, runtime, libraries, and system tools. It offers a sort of perfect packaging where everything is in an isolated environment from the host system. In this way, it eliminates the classic "it works on my machine" problem and ensures the application behaviour remains consistent, right from the developer's machine to the cloud-based production environment.

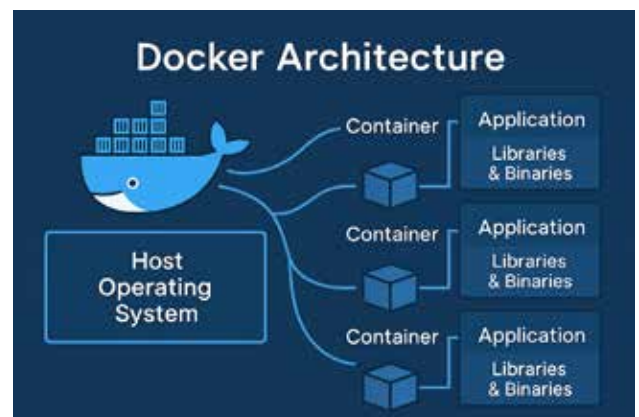


Figure 1: Docker architecture

Containerisation has been around for a while; however, Docker has made it simpler, faster, and developer friendly. It rose to stardom in 2013 and is today embedded in modern workflows, especially in areas such as: